



天津医科大学总医院
TIANJIN MEDICAL UNIVERSITY GENERAL HOSPITAL

静息态局部脑活动分析-实践课

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批量预处理，以 DPARF为例

计算过程: 打开MATLAB,
输入命令: DPARF/DPARF

命令行窗口

```
>> DPARF
```

简单来说，基础版里的选项一般是咱的必选项；高级版里面很多功能可以看成可选项

DPARF

Data Processing Assistant for Resting-State fMRI

Basic Edition **DPARF**

Working Directory: G:\2021training\exercise34

Participants:

Time Points: 0
TR (s): 2

☒ EPI DICOM to NIFTI ☒ Remove First 10 Time Points

☒ Slice Timing Slice Number: 33 Slice Order: [1 3 5 7 9 11 13] Reference Slice: 33

☒ Realign ☒ Normalize Bounding Box: [-90 -126 -72;90 90 108] Voxel Size: [3 3 3]

Normalize by using: ☒ EPI templates ☐ T1 image unified segmentation ☐ DARTEL

☒ Smooth FWHM: [4 4 4] ☐ Detrend

☒ Regress out nuisance covariates: Polynomial trend: 2

☒ Friston 24 head motion parameters ☐ Global mean signal

☒ White matter signal ☒ Cerebrospinal fluid signal ☐ Other covariates

☒ Default mask ☐ No mask ☐ User-defined mask Use Default Mask

☒ ALFF ☒ fALFF Band (Hz): 0.01 ~ 0.1

☒ Filter (Hz): 0.01 ~ 0.1

☐ ReHo Cluster: ☐ 7 voxels ☐ 19 voxels ☒ 27 voxels ☒ smReHo

☐ Extract ROI time courses ☐ Functional Connectivity Define ROI

Parallel Workers #: 0 Starting Directory Name: FunRaw

Help Save Load Utilities Quit Run

Data Processing Assistant for Resting-State fMRI

Advanced Edition DPARSFA

Working Directory:

Participants:

Time Points:

TR (s):

Template Para...

☒ EPI DICOM to NIFTI

☒ T1 DICOM to NIFTI

☐ BIDS to DPARSFA

☐ Apply Mats

☒ Remove First

Time Points

☒ Slice Timing

Slice Number:

Slice Order:

Reference Slice:

FieldMap Correction

☒ Realign

☐ Voxel-Specific Head Motion

☒ Reorient Fun*

☒ AutoMask

☐ Crop T1

☒ Reorient T1*

☒ Bet

☒ T1 Coreg to Fun

☐ Segment

☒ New Segment + DARTEL

Affine Regularisation in Segmentation: ☐ East Asian ☒ European

☒ Nuisance Covariates Regression

Polynomial trend:

Head Motion model: ☐ Rigid-body 6 ☐ Derivative 12

☒ Friston 24

☐ Voxel-specific 12

☐ Head motion scrubbing regressors

Nuisance regressors (WM, CSF, Global)

☐ Other covariates

☐ Add mean back

Filter (Hz): ~

☒ Normalize

Bounding Box:

Voxel Size:

☐ Normalize by using EPI templates

☐ Normalize by using T1 image unified segmentation

☒ Normalize by DARTEL

☐ Smooth

☐ Smooth by DARTEL

FWHM:

☒ Default mask

☐ No mask

☐ User-defined mask

Use Default Mask

☐ Warp Masks into Individual Space

☐ Detrend

☐ Nuisance Covariates Regression

☒ ALFF+fALFF

Band (Hz): ~

☒ Filter

☐ Scrubbing

☒ ReHo

Cluster: ☐ 7 ☐ 19 ☒ 27 voxels

☐ Smooth ReHo

☒ Degree Centrality

☐ Functional Connectivity

☒ Extract ROI time courses

Define ROI

☐ Define ROI Interactively*

☐ CWAS

☒ Normalize to Symmetric Template

☒ Smooth

☒ VMHC

☐ Normalize Derivatives

☒ Smooth Derivatives

Parallel Workers #:

Functional Sessions #:

Starting Directory Name:

Help

Save

Load

Utilities

Quit

Run

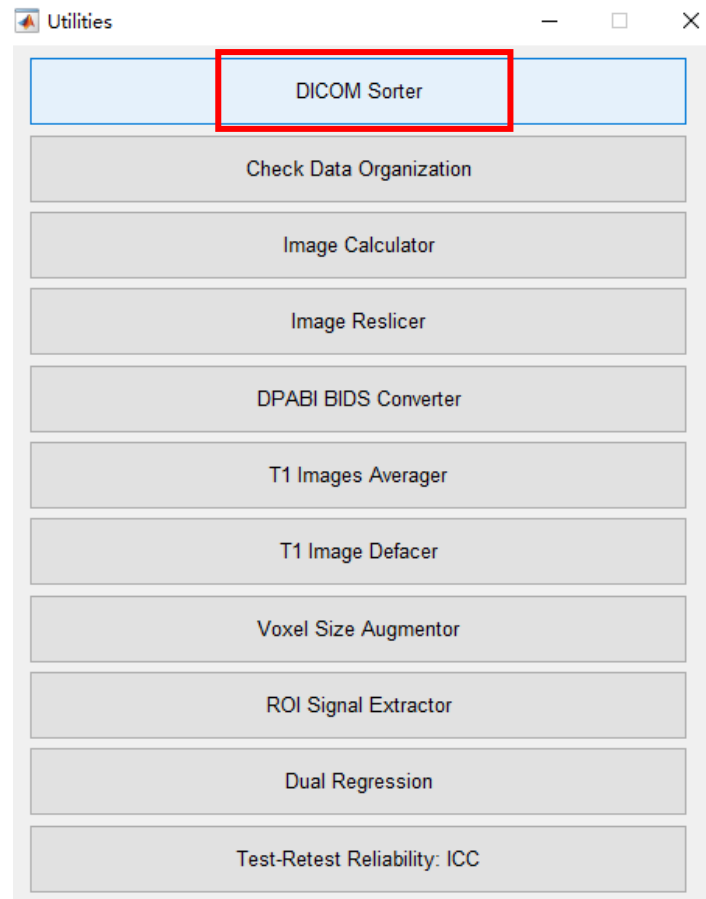
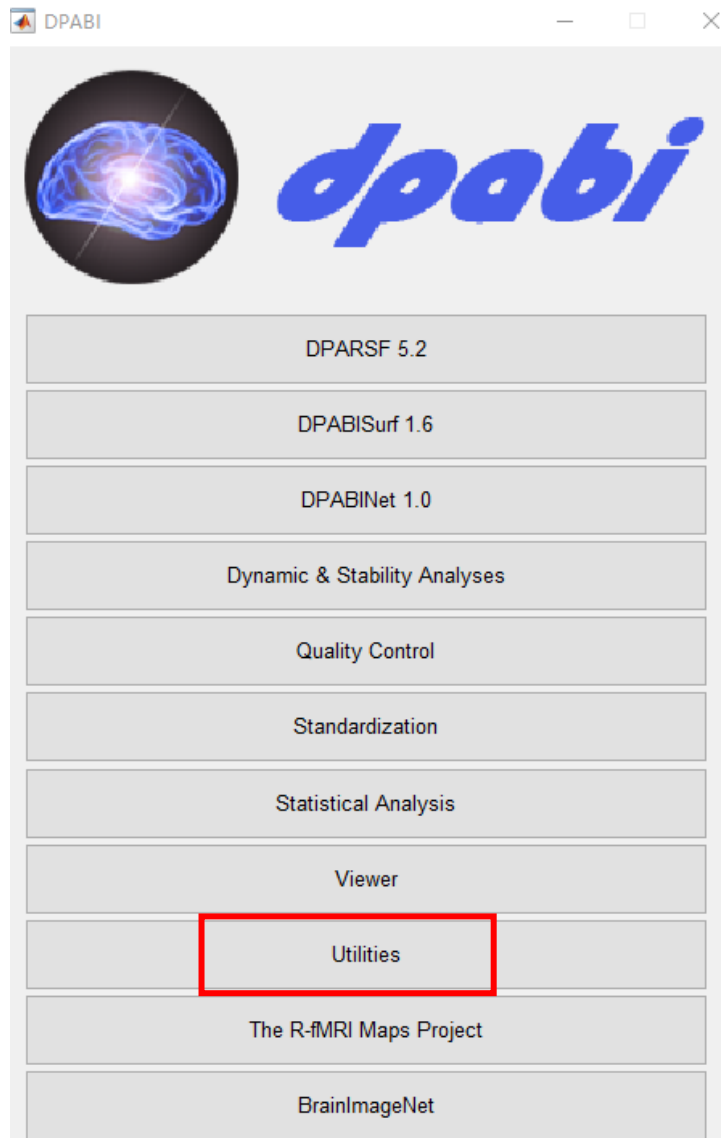
Command Window

New to MATLAB? Watch this [Video](#), see [Examples](#), or read [Getting Started](#).

>> DPARSFA

DPARSFA是高级版，
可用的功能更多。

数据整理





exercise34 > 111

- i2675897.MRDC.1
- i2675898.MRDC.2
- i2675899.MRDC.3
- i2675900.MRDC.4
- i2675901.MRDC.5
- i2675902.MRDC.6
- i2675903.MRDC.7
- i2675904.MRDC.8
- i2675905.MRDC.9
- i2675906.MRDC.10
- i2675907.MRDC.11
- i2675908.MRDC.12
- i2675909.MRDC.13
- i2675910.MRDC.14
- i2675911.MRDC.15
- i2675912.MRDC.16
- i2675913.MRDC.17
- i2675914.MRDC.18
- i2675915.MRDC.19
- i2675916.MRDC.20
- i2675917.MRDC.21
- i2675918.MRDC.22

常规 共享 安全 以前的版本 自定义

111

类型: 文件夹

位置: G:\2021training\exercise34

大小: 178 MB (187,263,322 字节)

占用空间: 203 MB (213,516,288 字节)

包含: 6,668 个文件, 0 个文件夹

创建时间: 2021年11月27日, 9:15:20

属性: ☒ 只读(仅应用于文件夹中的文件)(R) ☐ 隐藏(H) 高级(D)...

功能像6480个dcm文件
结构像188个dcm文件

DICOM Sorter

DICOM Directory

[6668] G:\2021training\exercise34\111

IMA, DCM, none

选数据总目录

Suffix Remove Add All

Output

Participant ID / Series Name ☐ Add Scanning Date ☐ Add Scanning Time

Layout: ~OutputDir\010001\CHIMGEN20151023

Output Dir: G:\2021training\exercise34\output

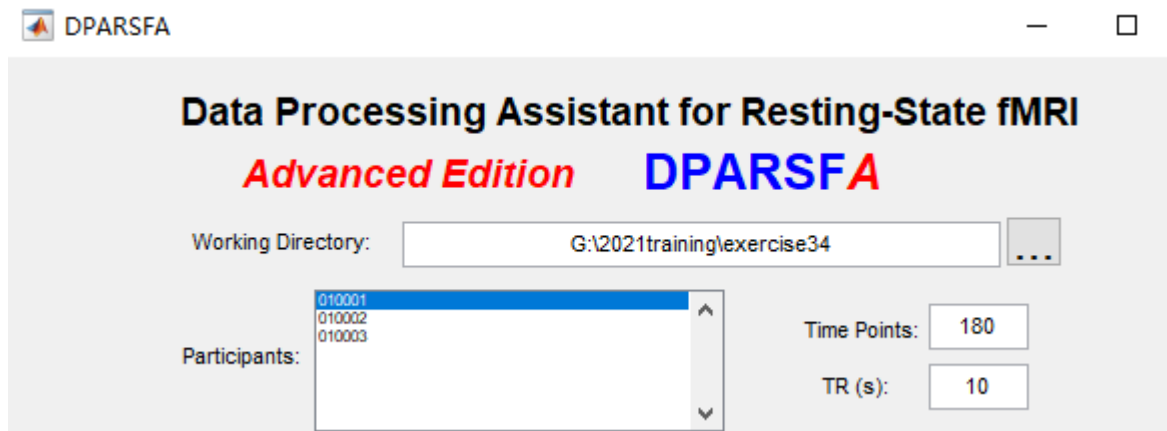
☐ Anonymize DICOM files Show another Demo DICOM! Sort

output > 010001

搜索"010001"

名称	修改日期	类型
0004_CHIMGEN20151023	2021/11/27 9:33	文件夹
0005_CHIMGEN20151023	2021/11/27 9:34	文件夹

数据导入

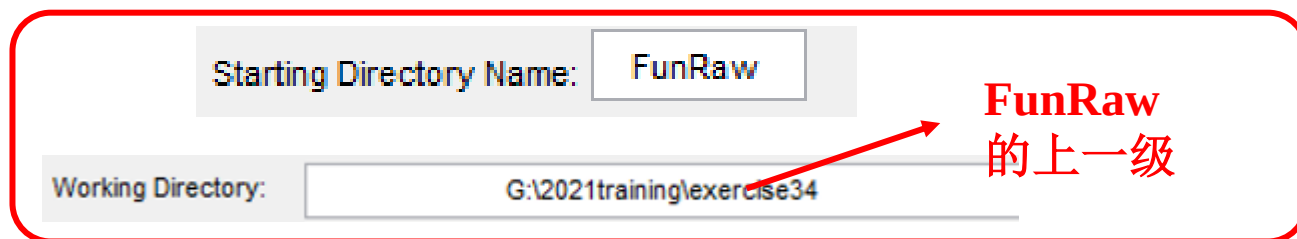


Working directory: 静息态fMRI数据预处理总目录，所有预处理都存在其下；

Time Points: 数据的时间点数； **TR:** 重复时间；

Participants: 导入需要处理的被试

一定要注意：



FunRaw
的上一级



工作路径：存放所有预处理过程的总文件夹

地磁盘 (G:) > 2021training > exercise34 >				▼	🔄	🔍 搜索"exercise34"
名称	修改日期	类型	大小			
FunImg	2021/11/26 15:45	文件夹				
FunImgA	2021/11/26 15:47	文件夹				
FunImgAR	2021/11/26 17:15	文件夹				
FunImgARC	2021/11/26 17:15	文件夹				
FunImgARCovs	2021/11/26 17:15	文件夹				
FunImgARW	2021/11/26 15:50	文件夹				
FunImgARWC	2021/11/26 15:50	文件夹				
FunImgARWCovs	2021/11/26 15:50	文件夹				
FunRaw	2021/7/28 16:06	文件夹				
Masks	2021/11/26 15:49	文件夹				
PicturesForChkNormalization	2021/11/26 15:49	文件夹				
QC	2021/11/26 15:48	文件夹				
RealignParameter	2021/11/26 15:48	文件夹				
ReorientMats	2021/11/26 15:48	文件夹				
T1Img	2021/11/26 15:30	文件夹				
T1Raw	2021/7/28 16:06	文件夹				
DPARSFA_AutoSave_2021_11_26_15_2...	2021/11/26 15:29	MATLAB Data	2 KB			
DPARSFA_AutoSave_2021_11_26_15_4...	2021/11/26 15:44	MATLAB Data	2 KB			
DPARSFA_AutoSave_2021_11_26_17_1...	2021/11/26 17:10	MATLAB Data	2 KB			
DPARSFA_AutoSave_2021_11_26_17_1...	2021/11/26 17:13	MATLAB Data	2 KB			

开始目录：你想从工作路径中的哪个文件夹开始处理



DPARSFA数据导入基本原则

如果是原始**DICOM**数据，需要在根目录下建立**FunRaw**文件夹，里面存放Sub 1, Sub 2,...,Sub n 等包含每个被试，在每个文件夹里面存放相应的**DICOM**数据。

如果是**nii**（或**hdr/img**或**nii.gz**）格式的数据，可以在根目录下建立**FunImg**文件夹，里面Sub 1, Sub 2,...,Sub n 文件夹用于存放相应的转换后的数据；**或者可以随意取名**（即文件夹不写为**FunImg**），只需在DPARSFA界面的右下角把Starting directory name改为相应的文件夹名字。

Tips 1: 先设置MATLAB的工作路径，方便后续操作。

Tips 2: DPARSFA的Working directory是FunRaw文件夹的上一级。



DPARSFA数据导入基本原则

如果是处理到某一步（同样使用DPARSFA处理，比如已经完成realign）的中间数据，需要在**Starting directory name**改为相应的名称（如**FunImgAR**）。

如果使用DPARSFA处理功能像的同时也加入结构像，需要建立**T1Raw**文件夹。如果是转换后的数据，则需要建立**T1Img**文件夹（不能是其他名字）。

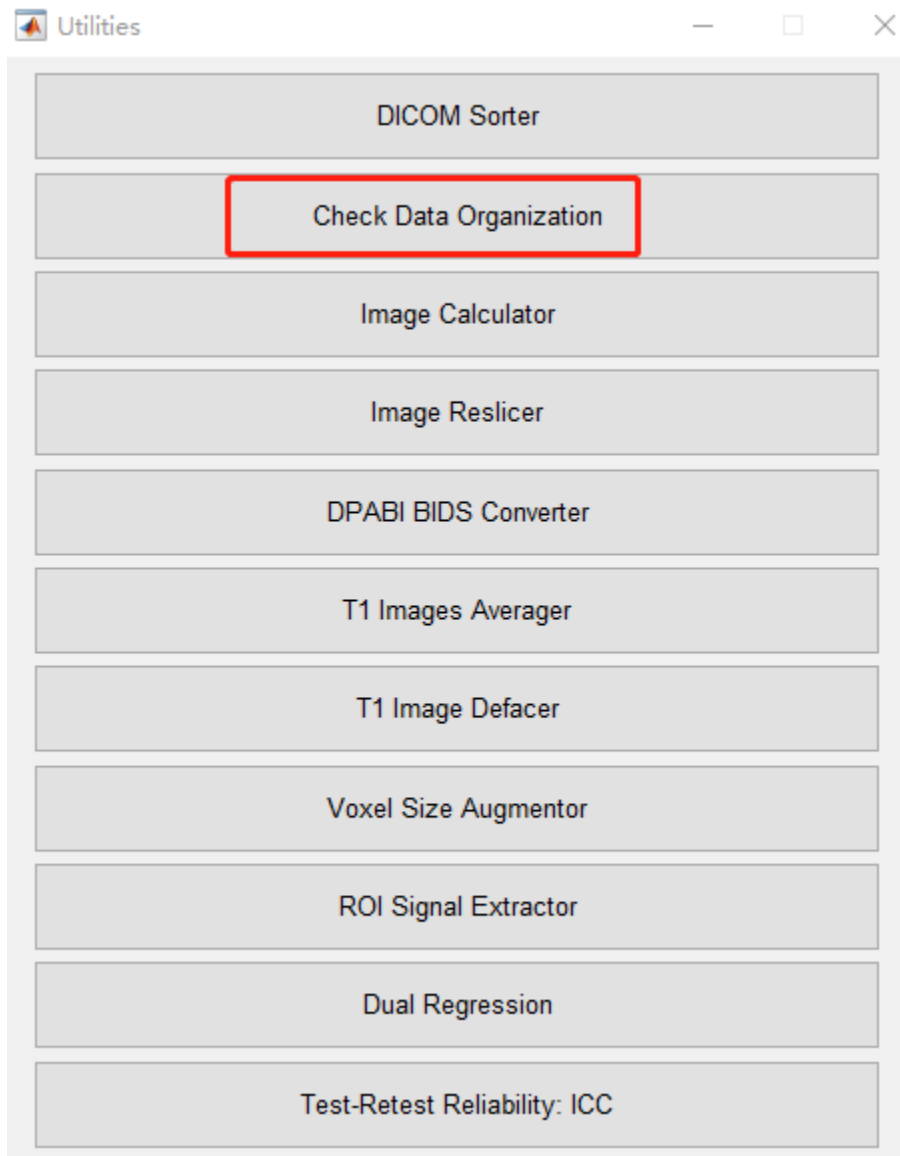
Tips 3: 文件夹名字不要有中文和空格（路径中也不能有）。

Tips 4: FunRaw, T1Raw区分大小写。

Tips 5: FunRaw, T1Raw文件夹里面对应的被试名字要一样。

Tips 6: FunRaw, T1Raw文件夹里面被试数量要一样。

数据检查



此电脑 > 本地磁盘 (G:) > 2021training > exercise34

名称	修改日期
FunRaw	2021,
T1Raw	2021,

Congratulations, data organization checking passed!!!

ans =

1

DPARSFA

Data Processing Assistant for Resting-State fMRI

Advanced Edition DPARSFA

Working Directory:

Participants:

Time Points:

TR (s):

Template Para... ☒ EPI DICOM to NIFTI ☒ T1 DICOM to NIFTI ☐ BIDS to DPARSF

Template Parameters

V5: Calculate in MNI Space (warp by DARTEL)
 Calculate in MNI Space (warp by DARTEL)
 Calculate in MNI Space (warp by information from unified segmentation)
 Calculate in MNI Space: TRADITIONAL order
 Calculate in Original Space (warp by DARTEL)
 Calculate in Original Space (warp by information from unified segmentation)
 Calculate ReHo and DC only (Smooth later)
 Intraoperative Processing
 Task fMRI data preprocessing
 VBM (New Segment and DARTEL)
 VBM (unified segmentation)
 Blank

Slice Order:

☐ Voxel-Specific Head Motion
☒ Bet ☒ T1 Coreg to Fun
☐ East Asian ☒ European
 Rigid-body 6 ☐ Derivative 12

☐ Filter (Hz): ~

☐ Warp Masks into Individual Space

☒ Default mask ☐ No mask ☐ User-defined mask

☐ Detrend ☐ Nuisance Covariates Regression ☒ ALFF+fALFF Band (Hz): ~ ☒ Filter

☐ Scrubbing ☒ ReHo Cluster ☐ 7 ☐ 19 ☒ 27 voxels ☐ Smooth ReHo ☒ Degree Centrality

☐ Functional Connectivity ☒ Extract ROI time courses ☐ Define ROI Interactively* ☐ CWAS

☒ Normalize to Symmetric Template ☒ Smooth ☒ VMHC ☐ Normalize Derivatives ☒ Smooth Derivatives

Parallel Workers #: Functional Sessions #: Starting Directory Name:

DPARSFA界面选项
的处理顺序:

- 从上到下
- 从左到右

DPARSFA

Data Processing Assistant for Resting-State fMRI

Advanced Edition DPARSFA

Working Directory: G:\2021training\exercise34

Participants:

Time Points: 0
TR (s): 0

Calculate in MN... ☒ EPI DICOM to NIFTI ☒ T1 DICOM to NIFTI ☐ BIDS to DPARSFA

Template Parameters

- ☐ V5: Calculate in MNI Space (warp by DARTEL)
- ☐ Calculate in MNI Space (warp by DARTEL)
- ☐ Calculate in MNI Space (warp by information from unified segmentation)
- ☒ Calculate in MNI Space: TRADITIONAL order**
- ☐ Calculate in Original Space (warp by DARTEL)
- ☐ Calculate in Original Space (warp by information from unified segmentation)
- ☐ Calculate ReHo and DC only (Smooth later)

Intraoperative Processing

Task fMRI data preprocessing

VBM (New Segment and DARTEL)

VBM (unified segmentation)

Blank

☒ Default mask ☐ No mask ☐ User-defined mask Use Default Mask ☐ Warp Masks into Individual Space

☒ Detrend ☒ Nuisance Covariates Regression ☒ ALFF+fALFF Band (Hz): 0.01 ~ 0.1 ☒ Filter

☐ Scrubbing ☐ ReHo Cluster ☐ 7 ☐ 19 ☒ 27 voxels ☐ Smooth ReHo ☐ Degree Centrality

☒ Functional Connectivity ☒ Extract ROI time courses Define ROI ☐ Define ROI Interactively* ☐ CWAS

☒ Normalize to Symmetric Template ☐ Smooth ☒ VMHC ☐ Normalize Derivatives ☐ Smooth Derivatives

Parallel Workers #: 0 Functional Sessions #: 1 Starting Directory Name: FunRaw

Help Save Load Utilities Quit Run

DPARSFA

Data Processing Assistant for Resting-State fMRI

Advanced Edition DPARSFA

Working Directory: G:\2021training\exercise34

Participants:

Time Points: 0
TR (s): 0

Calculate in MN... ☒ EPI DICOM to NIFTI ☒ T1 DICOM to NIFTI ☐ BIDS to DPARSFA

☐ Apply Mats ☒ Remove First 10 Time Points ☒ Slice Timing Slice Number: 0 Slice Order: [1 3 5 7 9 11]

Reference Slice: 0 FieldMap Correction ☒ Realign ☐ Voxel-Specific Head Motion

☒ Reorient Fun* ☒ AutoMask ☐ Crop T1 ☒ Reorient T1* ☒ Bet ☒ T1 Coreg to Fun

☐ Segment ☒ New Segment + DARTEL Affine Regularisation in Segmentation: ☐ East Asian ☒ European

☐ Nuisance Covariates Regression Polynomial trend: 1 Head Motion model: ☐ Rigid-body 6 ☐ Derivative 12

☒ Friston 24 ☐ Voxel-specific 12 ☐ Head motion scrubbing regressors

Nuisance regressors (WM, CSF, Global) ☐ Other covariates ☐ Add mean back ☐ Filter (Hz): 0.01 ~ 0.1

☒ Normalize Bounding Box: [-90 -126 -72; 90 90] Voxel Size: [3 3 3]

☐ Normalize by using EPI templates ☐ Normalize by using T1 image unified segmentation ☒ Normalize by DARTEL

☒ Smooth ☐ Smooth by DARTEL FWHM: [4 4 4]

☒ Default mask ☐ No mask ☐ User-defined mask Use Default Mask ☐ Warp Masks into Individual Space

☒ Detrend ☒ Nuisance Covariates Regression ☒ ALFF+fALFF Band (Hz): 0.01 ~ 0.1 ☒ Filter

☐ Scrubbing ☐ ReHo Cluster ☐ 7 ☐ 19 ☒ 27 voxels ☐ Smooth ReHo ☐ Degree Centrality

☒ Functional Connectivity ☒ Extract ROI time courses Define ROI ☐ Define ROI Interactively* ☐ CWAS

☒ Normalize to Symmetric Template ☐ Smooth ☒ VMHC ☐ Normalize Derivatives ☐ Smooth Derivatives

Parallel Workers #: 0 Functional Sessions #: 1 Starting Directory Name: FunRaw

Help Save Load Utilities Quit Run

静息态数据预处理一般步骤1:

标准空间回归协变量, 标准空间解算指标

DPARSFA

Data Processing Assistant for Resting-State fMRI

Advanced Edition DPARSFA

Working Directory:

Participants:

Time Points:

TR (s):

V5: Calculate in... ☒ EPI DICOM to NIFTI ☒ T1 DICOM to NIFTI ☐ BIDS to DPARSF

☐ Apply Mats ☒ Remove First Time Points ☒ Slice Timing Slice Number: Slice Order:

Reference Slice:

☒ Realign ☐ Voxel-Specific Head Motion

☒ Reorient Fun* ☒ AutoMask ☐ Crop T1 ☒ Reorient T1* ☒ Bet ☒ T1 Coreg to Fun

☐ Segment ☒ New Segment + DARTEL

☒ Nuisance Covariates Regression Head Motion model: ☐ Rigid-body 6 ☐ Derivative 12

☒ Friston 24 ☐ Voxel-specific 12 ☐ Head motion scrubbing regressors

☐ Other covariates ☐ Add mean back ☐ Filter (Hz): ~

☒ Normalize

☐ Normalize by using EPI templates ☐ Normalize by using T1 image unified segmentation ☒ Normalize by DARTEL

☐ Smooth ☐ Smooth by DARTEL

☒ Default mask ☐ No mask ☐ User-defined mask ☐ Warp Masks into Individual Space

☐ Detrend ☐ Nuisance Covariates Regression ☒ ALFF+fALFF ~ ☒ Filter

☐ Scrubbing ☒ ReHo ☐ Cluster ☐ 7 ☐ 19 ☒ 27 voxels ☐ Smooth ReHo ☒ Degree Centrality

☐ Functional Connectivity ☒ Extract ROI time courses ☐ Define ROI Interactively* ☐ CWAS

☒ Normalize to Symmetric Template ☒ Smooth ☒ VMHC ☐ Normalize Derivatives ☒ Smooth Derivatives

Parallel Workers #: Functional Sessions #: Starting Directory Name:

静息态数据预处理
一般步骤2:

个体空间回归协变量
标准空间解算指标

DPARSFA

Data Processing Assistant for Resting-State fMRI

Advanced Edition DPARSFA

Working Directory: G:\2021training\exercise34

Participants:

Time Points: 0
TR (s): 0

Calculate in Ori... ☒ EPI DICOM to NIFTI ☒ T1 DICOM to NIFTI ☐ BIDS to DPARSF

Template Parameters

- V5: Calculate in MNI Space (warp by DARTEL)
- Calculate in MNI Space (warp by DARTEL)
- Calculate in MNI Space (warp by information from unified segmentation)
- Calculate in MNI Space: TRADITIONAL order
- Calculate in Original Space (warp by DARTEL)**
- Calculate in Original Space (warp by information from unified segmentation)
- Calculate ReHo and DC only (Smooth later)
- Intraoperative Processing
- Task fMRI data preprocessing
- VBM (New Segment and DARTEL)
- VBM (unified segmentation)
- Blank

☒ Default mask ☐ No mask ☐ User-defined mask Use Default Mask ☒ Warp Masks into Individual Space

☐ Detrend ☐ Nuisance Covariates Regression ☒ ALFF+fALFF Band (Hz): 0.01 ~ 0.1 ☒ Filter

☐ Scrubbing ☒ ReHo Cluster 7 19 27 voxels ☐ Smooth ReHo ☐ Degree Centrality

☒ Functional Connectivity ☒ Extract ROI time courses Define ROI ☐ Define ROI Interactively* ☐ CWAS

☐ Normalize to Symmetric Template ☐ Smooth ☐ VMHC ☒ Normalize Derivatives ☒ Smooth Derivatives

Parallel Workers #: 0 Functional Sessions #: 1 Starting Directory Name: FunRaw

Help Save Load Utilities Quit Run

DPARSFA

Data Processing Assistant for Resting-State fMRI

Advanced Edition DPARSFA

Working Directory: G:\2021training\exercise34

Participants:

Time Points: 0
TR (s): 0

Calculate in Ori... ☒ EPI DICOM to NIFTI ☒ T1 DICOM to NIFTI ☐ BIDS to DPARSF

☐ Apply Mats ☒ Remove First 10 Time Points ☒ Slice Timing Slice Number: 0 Slice Order: [1 3 5 7 9 11]

Reference Slice: 0 ☐ FieldMap Correction ☒ Realign ☐ Voxel-Specific Head Motion

☒ Reorient Fun* ☒ AutoMask ☐ Crop T1 ☒ Reorient T1* ☒ Bet ☒ T1 Coreg to Fun

☐ Segment ☒ New Segment + DARTEL Affine Regularisation in Segmentation: ☐ East Asian ☒ European

☒ Nuisance Covariates Regression Polynomial trend: 1 Head Motion model: ☐ Rigid-body 6 ☐ Derivative 12

☒ Friston 24 ☐ Voxel-specific 12 ☐ Head motion scrubbing regressors

Nuisance regressors (WM, CSF, Global) ☐ Other covariates ☐ Add mean back ☐ Filter (Hz): 0.01 ~ 0.1

☐ Normalize Bounding Box: [-90 -126 -72:90 90] Voxel Size: [3 3 3]

☐ Normalize by using EPI templates ☐ Normalize by using T1 image unified segmentation ☒ Normalize by DARTEL

☐ Smooth ☐ Smooth by DARTEL FWHM: [4 4 4]

☒ Default mask ☐ No mask ☐ User-defined mask Use Default Mask ☒ Warp Masks into Individual Space

☐ Detrend ☐ Nuisance Covariates Regression ☒ ALFF+fALFF Band (Hz): 0.01 ~ 0.1 ☒ Filter

☐ Scrubbing ☒ ReHo Cluster 7 19 27 voxels ☐ Smooth ReHo ☐ Degree Centrality

☒ Functional Connectivity ☒ Extract ROI time courses Define ROI ☐ Define ROI Interactively* ☐ CWAS

☐ Normalize to Symmetric Template ☐ Smooth ☐ VMHC ☒ Normalize Derivatives ☒ Smooth Derivatives

Parallel Workers #: 0 Functional Sessions #: 1 Starting Directory Name: FunRaw

Help Save Load Utilities Quit Run

静息态数据预处理一般步骤3:

个体空间回归协变量, 个体空间解算指标



FunImg	2021/10/7 星期四 10:45	文件夹
FunImgA	2021/10/7 星期四 10:50	文件夹
FunImgAR	2021/10/7 星期四 10:55	文件夹
FunImgARW	2021/10/7 星期四 14:50	文件夹
FunImgARWD	2021/10/7 星期四 14:56	文件夹
FunImgARWDC	2021/10/7 星期四 14:58	文件夹
FunImgARWDCF	2021/10/7 星期四 14:58	文件夹
FunImgARWDCFB	2021/10/7 星期四 15:00	文件夹
FunImgARWDCFBsym	2021/10/7 星期四 15:29	文件夹
FunImgARWDCFBsymS	2021/10/7 星期四 15:34	文件夹
FunImgARWDCovs	2021/10/7 星期四 14:56	文件夹
FunImgARWS	2021/10/7 星期四 11:28	文件夹
FunImgARWSD	2021/10/7 星期四 11:35	文件夹
FunImgARWSDC	2021/10/7 星期四 11:37	文件夹
FunImgARWSDCF	2021/10/7 星期四 11:37	文件夹
FunImgARWSDCFB	2021/10/7 星期四 11:38	文件夹
FunImgARWSDCFBsym	2021/10/7 星期四 11:46	文件夹
FunImgARWSDCovs	2021/10/7 星期四 11:35	文件夹
FunRaw	2021/7/25 星期日 11:01	文件夹
Masks	2021/10/7 星期四 11:38	文件夹
PicturesForChkNormalization	2021/10/7 星期四 11:19	文件夹
QC	2021/10/7 星期四 10:58	文件夹
RealignParameter	2021/10/7 星期四 10:55	文件夹
ReorientMats	2021/10/7 星期四 10:58	文件夹
Results	2021/10/7 星期四 15:34	文件夹
SymmetricGroupT1MeanTemplate	2021/10/7 星期四 15:37	文件夹
T1Img	2021/10/7 星期四 10:45	文件夹
T1ImgBet	2021/10/7 星期四 11:00	文件夹
T1ImgCoreg	2021/10/7 星期四 11:00	文件夹
T1ImgSegment	2021/10/7 星期四 11:02	文件夹
T1Raw	2021/7/25 星期日 11:02	文件夹
DPARSFA_AutoSave_2021_10_7_10_42.mat	2021/10/7 星期四 10:42	MAT 文件

Abbreviations:

A -- Slice timing

R -- Realign

W -- Normalize

S -- Smooth

D -- Detrend

C -- Regress covariate

F -- Filter

B -- Scrub

DPARSFA

Data Processing Assistant for Resting-State fMRI

Advanced Edition DPARSFA

Working Directory: G:\2021training\exercise32

Participants: 010001, 010002, 010003

Time Points: 170
TR (s): 2

Template Para... ☐ EPI DICOM to NIFTI ☐ T1 DICOM to NIFTI ☐ BIDS to DPARSFA

☐ Apply Mats ☐ Remove First 0 Time Points ☐ Slice Timing Slice Number: 36 Slice Order: [1 3 5 7 9 11]
Reference Slice: 35 ☐ FieldMap Correction ☐ Realign ☐ Voxel-Specific Head Motion

☐ Reorient Fun* ☐ AutoMask ☐ Crop T1 ☐ Reorient T1* ☐ Bet ☐ T1 Coreg to Fun

☐ Segment ☐ New Segment + DARTEL

☐ Nuisance Covariates Regression Polynomial trend: 1 Head Motion model: ☐ Rigid-body 6 ☐ Derivative 12
☒ Friston 24 ☐ Voxel-specific 12 ☐ Head motion scrubbing regressors

☐ Nuisance regressors (WM, CSF, Global) ☐ Other covariates ☐ Add mean back ☐ Filter (Hz): 0.01 ~ 0.1

☐ Normalize Bounding Box: [-90 -126 -72; 90 90] Voxel Size: [3 3 3]
☐ Normalize by using EPI templates ☐ Normalize by using T1 image unified segmentation ☐ Normalize by DARTEL

☐ Smooth ☐ Smooth by DARTEL FWHM: [4 4 4]

☒ Default mask ☐ No mask ☐ User-defined mask Use Default Mask ☐ Warp Masks into Individual Space

☐ Detrend ☒ Nuisance Covariates Regression ☐ ALFF+fALFF Band (Hz): 0.01 ~ 0.1 ☐ Filter

☐ Scrubbing ☐ ReHo Cluster ☐ 7 ☐ 19 ☒ 27 voxels ☐ Smooth ReHo ☐ Degree Centrality

☐ Functional Connectivity ☐ Extract ROI time courses Define ROI ☐ Define ROI Interactively* ☐ CWAS

☐ Normalize to Symmetric Template ☐ Smooth ☐ VMHC ☐ Normalize Derivatives ☐ Smooth Derivatives

Parallel Workers #: 0 Functional Sessions #: 1 Starting Directory Name: FunImgARW

Help Save Load Utilities Quit Run

FunImg	2021/11/26 15:52	文
FunImgA	2021/11/26 15:52	文
FunImgAR	2021/11/26 15:52	文
FunImgARW	2021/11/26 15:56	文
FunImgARWC	2021/11/26 15:56	文
FunImgARWCovs	2021/11/26 15:56	文
Masks	2021/11/26 15:52	文
PicturesForChkNormalization	2021/11/26 15:52	文
QC	2021/11/26 15:52	文
RealignParameter	2021/11/26 15:52	文
ReorientMats	2021/11/26 15:52	文
DPARSFA_AutoSave_2021_11_26_15_55.mat	2021/11/26 15:55	M
DPARSFA AutoSave 2021 11 28 9 34.mat	2021/11/28 9:34	M

每点一次run，就会自动存储一个配置文件

实践:

静息态数据预处理一般步骤1:

DPARSFA

Data Processing Assistant for Resting-State fMRI Advanced Edition DPARSFA

Working Directory: G:\2021training\exercise34

Participants:

Time Points:
TR (s):

Calculate in MN... ☒ EPI DICOM to NIFTI ☒ T1 DICOM to NIFTI ☐ BIDS to DPARSF

☐ Apply Mats ☒ Remove First Time Points ☒ Slice Timing Slice Number: Slice Order:
Reference Slice: FieldMap Correction ☒ Realign ☐ Voxel-Specific Head Motion

☒ Reorient Fun* ☒ AutoMask ☐ Crop T1 ☒ Reorient T1* ☒ Bet ☒ T1 Coreg to Fun

☐ Segment ☒ New Segment + DARTEL Affine Regularisation in Segmentation: ☐ East Asian ☒ European

☐ Nuisance Covariates Regression Polynomial trend: Head Motion model: ☐ Rigid-body 6 ☐ Derivative 12

☒ Friston 24 ☐ Voxel-specific 12 ☐ Head motion scrubbing regressors

Nuisance regressors (WM, CSF, Global) ☐ Other covariates ☐ Add mean back Filter (Hz): ~

☒ Normalize Bounding Box: Voxel Size:
☐ Normalize by using EPI templates ☐ Normalize by using T1 image unified segmentation ☒ Normalize by DARTEL

☒ Smooth ☐ Smooth by DARTEL FWHM:

☒ Default mask ☐ No mask ☐ User-defined mask Use Default Mask ☐ Warp Masks into Individual Space

☒ Detrend ☒ Nuisance Covariates Regression ☒ ALFF+fALFF Band (Hz): ~ ☒ Filter

☐ Scrubbing ☐ ReHo Cluster ☐ 7 ☐ 19 ☒ 27 voxels ☐ Smooth ReHo ☐ Degree Centrality

☒ Functional Connectivity ☒ Extract ROI time courses Define ROI ☐ Define ROI Interactively* ☐ CWAS

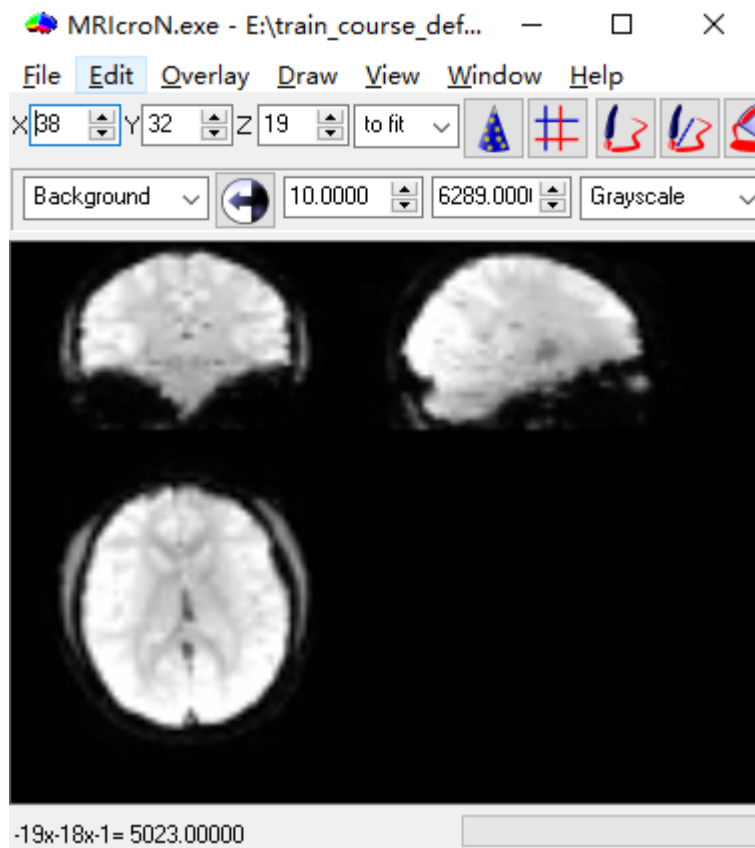
☒ Normalize to Symmetric Template ☐ Smooth ☒ VMHC ☐ Normalize Derivatives ☐ Smooth Derivatives

Parallel Workers #: Functional Sessions #: Starting Directory Name:

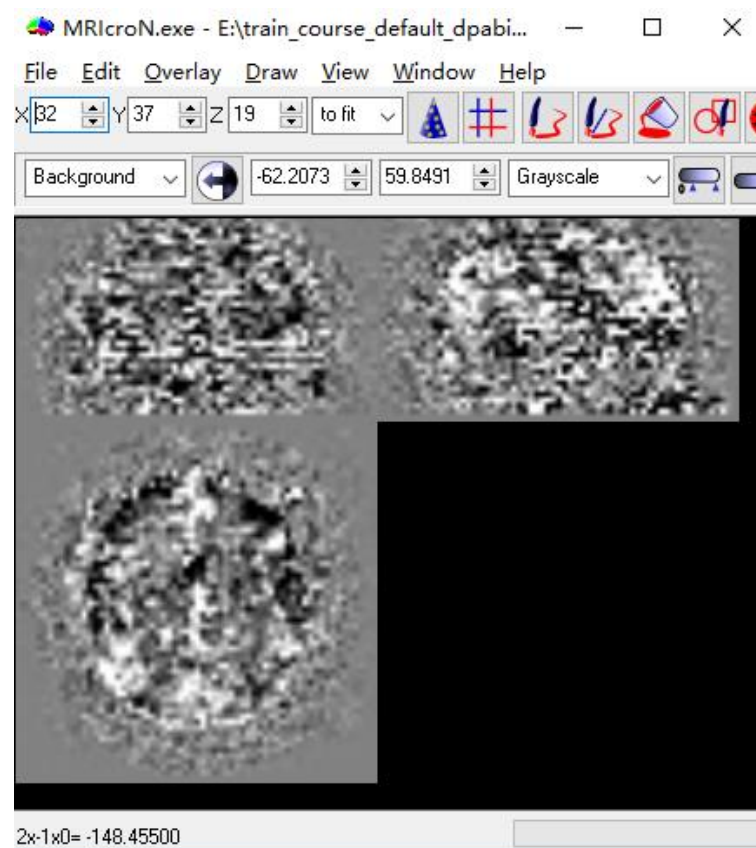
Help Save Load Utilities Quit Run

思考题:

1. 为什么做完回归的图像，会如此的不清楚？看不出脑结构？



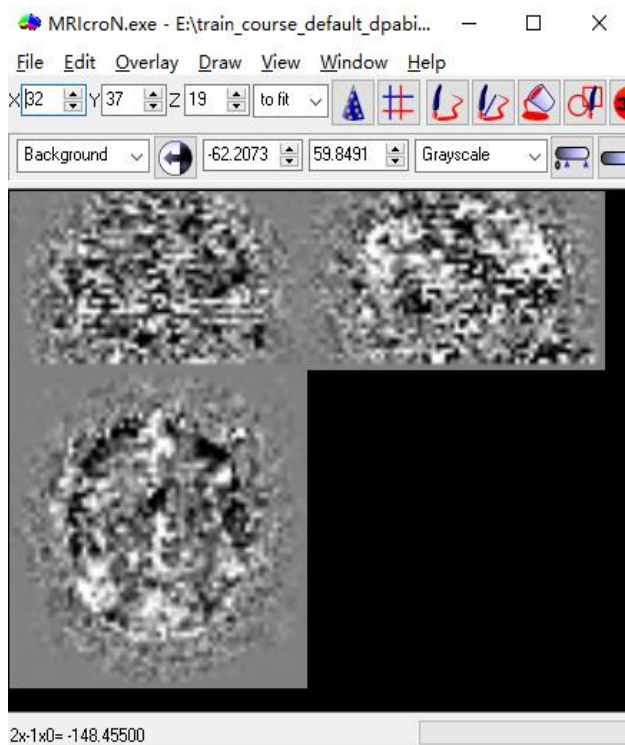
FunImgAR



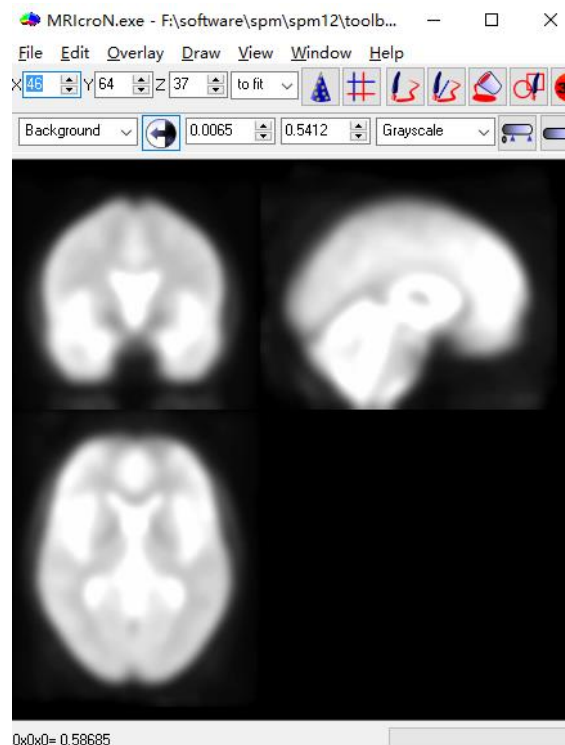
FunImgARC

思考题:

2. 如果先回归再配准，还能不能配准到EPI模板？



FunImgARC



EPI template

思考题:

3. DPARSFA下面这样选择有没有问题？

DPARSFA

Data Processing Assistant for Resting-State fMRI
Advanced Edition DPARSFA

Working Directory: E:\train_course_traditional

Participants: 010001, 010002, 010003, 010004, 010005, 010006, 010007

Time Points: 180
TR (s): 2

Blank ☒ EPIDICOM to NIFTI ☐ Apply Mats ☒ Remove First 10 Time Points ☒ Slice Timing

Slice 36 Slice Order: [1 3 5 7 9 11] Reference Slice: 35 ☒ Realign ☐ Voxel-Specific Head Motion

☒ Reorient Fun* ☐ AutoMask ☐ T1 DICOM to NIFTI ☐ Crop T1 ☐ Reorient T1* ☐ Bet ☐ T1 Coreg to Fun

☐ Segment ☐ New Segment + DARTEL

☒ Nuisance Covariates Regression Polynomial trend: 1 Head Motion model: ☐ Rigid-body 6 ☐ Derivative 12

☒ Friston 24 ☐ Voxel-specific 12 ☐ Head motion scrubbing regressors

Nuisance regressors (WM, CSF, Global) ☐ Other covariates ☐ Add mean back ☐ Filter (Hz): 0.01 ~ 0.1

没有T1像，勾上面的regress

Data Processing Assistant for Resting-State fMRI

Advanced Edition DPARSFA

Working Directory: F:\DATA\train_course_traditional_dpabi_17_scrub ...

Participants:

- 010001
- 010002
- 010003
- 010004
- 010005
- 010006
- 010007

Time Points: 180

TR (s): 2

Blank ☒ EPI DICOM to NIFTI ☐ Apply Mats ☒ Remove First 10 Time Points ☒ Slice Timing

Slice 36 Slice Order: [1 3 5 7 9 11] Reference Slice: 36 ☒ Realign ☐ Voxel-Specific Head Motion

☐ Reorient Fun* ☐ AutoMask ☐ T1 DICOM to NIFTI ☐ Crop T1 ☐ Reorient T1* ☐ Bet ☐ T1 Coreg to Fun

☐ Segment ☐ New Segment + DARTEL

☐ Nuisance Covariates Regression Polynomial trend: 1 Head Motion model: ☐ Rigid-body 6 ☐ Derivative 12

☒ Friston 24 ☐ Voxel-specific 12 ☐ Head motion scrubbing regressors

Nuisance regressors (WM, CSF, Global) ☐ Other covariates ☐ Add mean back ☐ Filter (Hz): 0.01 ~ 0.1

☐ Normalize Bounding Box: [-90 -126 -72; 90 90] Voxel Size: [3 3 3]

☐ Normalize by using EPI templates ☐ Normalize by using T1 image unified segmentation ☐ Normalize by DARTEL

☐ Smooth ☐ Smooth by DARTEL FWHM: [4 4 4]

☒ Default mask ☐ No mask ☐ User-defined mask Use Default Mask ☐ Warp Masks into Individual Space

☐ Detrend ☒ Nuisance Covariates Regression ☐ ALFF+fALFF Band (Hz): 0.01 ~ 0.1 ☐ Filter

☐ Scrubbing ☐ ReHo Cluster: ☐ 7 ☐ 19 ☒ 27 voxels ☐ Smooth ReHo ☐ Degree Centrality

☐ Functional Connectivity ☐ Extract ROI time courses Define ROI ☐ Define ROI Interactively* ☐ CWAS

☐ Normalize to Symmetric Temp.. ☐ Smooth ☐ VMHC ☐ Normalize Derivatives ☐ Smooth Derivatives

Parallel Workers #: 0 Functional Sessions #: 1 Starting Directory Name: FunRaw

Help

Save

Load

Utilities

Quit

Run

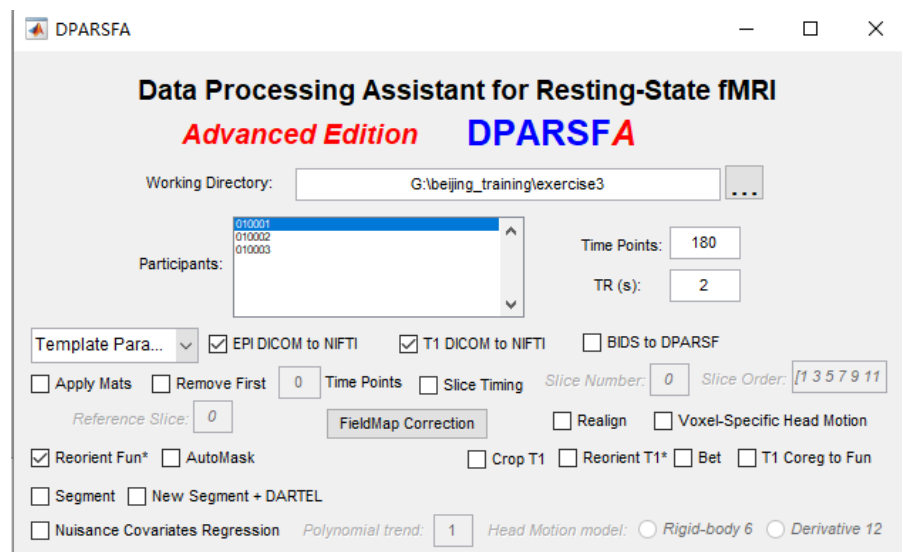
思考题:

4. 这样呢?

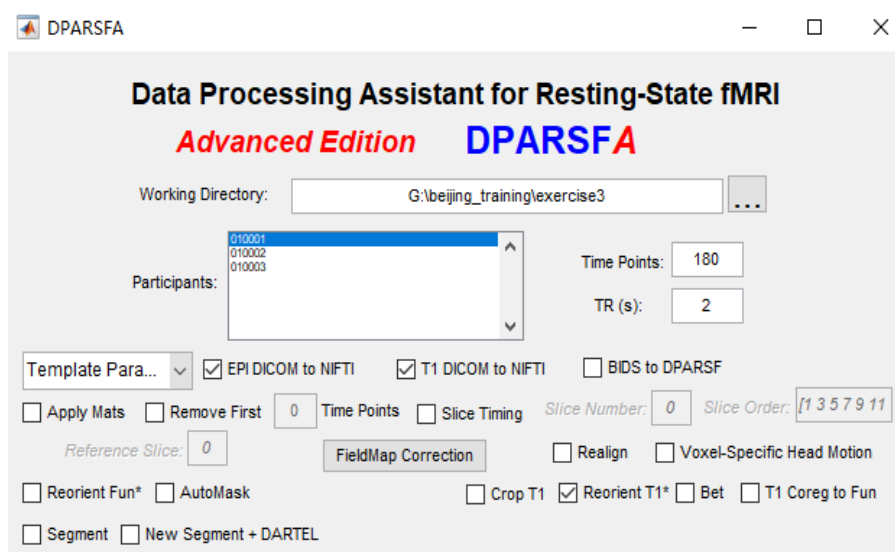
没做配准, 勾下面的
回归协变量

思考题:

5. DPARSFA是否可以只reorient单一模态？



Reorient功能像



Reorient结构像

Data Processing Assistant for Resting-State fMRI

Advanced Edition DPARSFA

Working Directory: G:\2021training\exercise34

Participants: 010001
010002
010003

Time Points: 180

TR (s): 2

Blank

☐ EPI DICOM to NIFTI

☒ T1 DICOM to NIFTI

☐ BIDS to DPARSFA

☐ Apply Mats

☐ Remove First 0

Time Points

☐ Slice Timing

Slice Number: 0

Slice Order: [1 3 5 7 9 11]

Reference Slice: 0

FieldMap Correction

☐ Realign

☐ Voxel-Specific Head Motion

☐ Reorient Fun* ☐ AutoMask

☐ Crop T1

☒ Reorient T1*

☒ Bet

☐ T1 Coreg to Fun

☐ Segment ☐ New Segment + DARTEL

☐ Nuisance Covariates Regression

Polynomial trend: 1

Head Motion model: ☐ Rigid-body 6 ☐ Derivative 12

☒ Friston 24 ☐ Voxel-specific 12 ☐ Head motion scrubbing regressors

Nuisance regressors (WM, CSF, Global)

☐ Other covariates

☐ Add mean back

☐ Filter (Hz): 0.01 ~ 0.1

☐ Normalize Bounding Box: [-90 -126 -72; 90 90]

Voxel Size: [3 3 3]

☐ Normalize by using EPI templates ☐ Normalize by using T1 image unified segmentation ☐ Normalize by DARTEL

☐ Smooth ☐ Smooth by DARTEL

FWHM: [4 4 4]

☒ Default mask ☐ No mask ☐ User-defined mask

Use Default Mask

☐ Warp Masks into Individual Space

☐ Detrend

☐ Nuisance Covariates Regression

☐ ALFF+fALFF Band (Hz): 0.01 ~ 0.1

☐ Filter

☐ Scrubbing

☐ ReHo

Cluster ☐ 7 ☐ 19 ☒ 27 voxels

☐ Smooth ReHo

☐ Degree Centrality

☐ Functional Connectivity

☐ Extract ROI time courses

Define ROI

☐ Define ROI Interactively*

☐ CWAS

☐ Normalize to Symmetric Template

☐ Smooth

☐ VMHC

☐ Normalize Derivatives

☐ Smooth Derivatives

Parallel Workers #: 0

Functional Sessions #: 1

Starting Directory Name: FunRaw

Help

Save

Load

Utilities

Quit

Run

思考题:

6. 如果想用 DPARSFA 只做 T1 像的处理, 该如何进行?



思考题:

7. 如何看出是否回归了global signal?

名称	修改日期
 FunImg	2021/7/25 16:48
 FunImgA	2021/7/25 16:49
 FunImgAR	2021/7/25 16:51
 FunImgARW	2021/7/25 18:03
 FunImgARWD	2021/7/25 18:04
 FunImgARWDC	2021/7/25 18:04
 FunImgARWDCF	2021/7/25 18:04
 FunImgARWDCovs	2021/7/25 18:04
 FunImgARWS	2021/7/25 17:54
 FunImgARWSD	2021/7/25 17:55
 FunImgARWSDC	2021/7/25 17:56
 FunImgARWSDCF	2021/7/25 17:56
 FunImgARWSDCovs	2021/7/25 17:55
 FunRaw	2021/7/25 16:34
 Masks	2021/7/26 8:51
 PicturesForChkNormalization	2021/7/25 17:51
 QC	2021/7/25 16:53
 RealignParameter	2021/7/25 16:51
 ReorientMats	2021/7/25 16:53
 Results	2021/7/26 8:51
 T1Img	2021/7/25 16:48

思考题:

8. DPARSFA下面这样选择有没有问题?

DPARSFA

Data Processing Assistant for Resting-State fMRI
Advanced Edition DPARSFA

Working Directory: F:\DATA\train_course

Participants: 010001, 010002, 010003, 010004, 010005, 010006, 010007

Time Points: 180
TR (s): 2

Calculate in MN... ☒ EPI DICOM to NIFTI ☐ Apply Mats ☒ Remove First 10 Time Points ☒ Slice Timing

Slice 36 Slice Order: [1 3 5 7 9 11] Reference Slice: 35 ☒ Realign ☐ Voxel-Specific Head Motion

☒ Reorient Fun* ☒ AutoMask ☒ T1 DICOM to NIFTI ☐ Crop T1 ☒ Reorient T1* ☒ Bet ☒ T1 Coreg to Fun

☐ Segment ☒ New Segment + DARTEL Affine Regularisation in Segmentation: ☐ East Asian ☒ European

☐ Nuisance Covariates Regression Polynomial trend: 1 Head Motion model: ☐ Rigid-body 6 ☐ Derivative 12

☒ Friston 24 ☐ Voxel-specific 12 ☐ Head motion scrubbing regressors

Nuisance regressors (WM, CSF, Global) ☐ Other covariates ☐ Add mean back ☐ Filter (Hz): 0.01 ~ 0.1

☒ Normalize Bounding Box: [-90 -126 -72; 90 90] Voxel Size: [3 3 3]

☐ Normalize by using EPI templates ☒ Normalize by using T1 image unified segmentation ☐ Normalize by DARTEL

☒ Smooth ☐ Smooth by DARTEL FWHM: [4 4 4]

前面选择的New segment + DARTEL，后面选的T1分割配准

思考题： 9. 这样呢？

DPARSFA

Data Processing Assistant for Resting-State fMRI Advanced Edition DPARSFA

Working Directory: F:\DATA\train_course

Participants: 010001, 010002, 010003, 010004, 010005, 010006, 010007

Time Points: 180
TR (s): 2

Calculate in MN... ☒ EPI DICOM to NIFTI ☐ Apply Mats ☒ Remove First 10 Time Points ☒ Slice Timing

Slice 36 Slice Order: [1 3 5 7 9 11] Reference Slice: 35 ☒ Realign ☐ Voxel-Specific Head Motion

☒ Reorient Fun* ☒ AutoMask ☒ T1 DICOM to NIFTI ☐ Crop T1 ☒ Reorient T1* ☒ Bet ☒ T1 Coreg to Fun

☒ Segment ☐ New Segment + DARTEL Affine Regularisation in Segmentation: ☐ East Asian ☒ European

☐ Nuisance Covariates Regression Polynomial trend: 1 Head Motion model: ☐ Rigid-body 6 ☐ Derivative 12

☒ Friston 24 ☐ Voxel-specific 12 ☐ Head motion scrubbing regressors

Nuisance regressors (WM, CSF, Global) ☐ Other covariates ☐ Add mean back ☐ Filter (Hz): 0.01 ~ 0.1

☒ Normalize Bounding Box: [-90 -126 -72; 90 90] Voxel Size: [3 3 3]

☐ Normalize by using EPI templates ☐ Normalize by using T1 image unified segmentation ☒ Normalize by DARTEL

前面选择的segment，后面选的DARTEL配准

思考题： 10. 这样？

DPARSFA

Data Processing Assistant for Resting-State fMRI

Advanced Edition DPARSFA

Working Directory: F:\DATA\dpabi\test

Participant: 010001, 010002, 010003

Time Points: 0
TR (s): 2

Template Para... ☐ EPI DICOM to NIFTI ☐ Apply Mats ☐ Remove First 0 Time Points ☒ Slice Timing

Slice: 36 Slice Order: [1 3 5 7 9 1] Reference Slice: 35 ☒ Realign ☐ Voxel-Specific Head Motion

☐ Reorient Fun* ☐ AutoMask ☐ T1 DICOM to NIFTI ☐ Crop T1 ☐ Reorient T1* ☐ Bet ☐ T1 Coreg to Fun

☒ Segment ☐ New Segment + DARTEL Affine Regularisation in ☐ East Asian ☒ European

☒ Nuisance Covariates Regression Polynomial: 1 Head Motion ☐ Rigid-body 6 ☐ Derivative 12

☒ Friston 24 ☐ Voxel-specific 12 ☐ Head motion scrubbing regres...

Nuisance regressors (WM, CSF, Global) ☐ Other covariates ☐ Add mean back ☐ Filter (Hz): 0.01 ~ 0.1

☒ Normalize Bounding Box: [-90 -126 -72;90] Voxel Size: [3 3 3]

☒ Normalize by using EPI templates ☐ Normalize by using T1 image unified segmentation ☐ Normalize by DARTEL

☐ Smooth ☐ Smooth by DARTEL FWHM: [4 4 4]

☒ Default mask ☐ No mask ☐ User-defined mask Use Default Mas ☒ Warp Masks into Individual Space

☐ Detrend ☐ Nuisance Covariates Regression ☒ ALFF+fALFF Band (Hz): 0.01 ~ 0.1 ☒ Filter

☐ Scrubbing ☒ ReHo Cluster ☐ 7 ☐ 19 ☒ 27 voxels ☐ Smooth ReHo ☐ Degree Centrality

☒ Functional Connectivity ☒ Extract ROI time courses Define ROI ☐ Define ROI Interactively* ☐ CWAS

☐ Normalize to Symmetric Temp.. ☐ Smooth ☐ VMHC ☐ Normalize Derivatives ☒ Smooth Derivatives

Parallel Workers #: 0 Functional Sessions #: 1 Starting Directory Name: FunImg

Help Save Load Utilities Quit Run

前面预处理将数据配准到了标准空间，后面解算指标之前勾上了 Warp masks into individual space

课后作业:

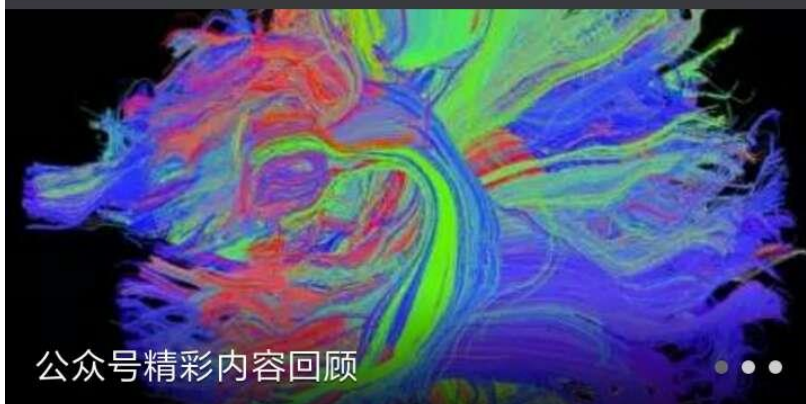
1. 使用mricro和matlab查看扫描参数;
2. 完成数据预处理一般步骤2流程图的预处理, 得到ReHo、ALFF和fALFF指标 (要求: 使用EPI配准, ALFF先平滑, ReHo后平滑)
3. 完成数据预处理一般步骤3流程图的预处理, 得到ReHo、ALFF和fALFF指标 (要求: 使用结合T1像的New segment + DARTEL的配准方法)
4. 练习自己通过头动 (rp文件) 画出头动曲线;
5. 解决思考题。



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公众号精彩内容回顾

影像遗传

影像方法

其它



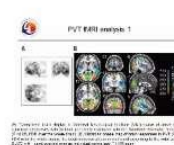
Nature：深度学习预测婴儿自闭症
 Journal Club 孤独症(autism)，又称自闭症或孤独性障碍(autistic disorder)



基于功能连接影像学标志物定义...
 在现代医学中，生物标志物的应用发展迅速，但是在精神领域中仍然面临着很大问...



Global signal，到底该不该去除？
 Journal Club 全脑均值是大脑中所有体素时间序列的平均信号强度，它既包括神经活...



昼夜节律和睡眠剥夺对脑反应的...
 众所周知，人们的行为认知表现受到昼夜节律和睡眠剥夺的影响。虽然以前研究报...

谢谢大家！